Meta-Analysis using COMETS-Analytics

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COMETS Analytics Design Facilitates Meta-Analyses

- Standardized definitions of covariables and models (by lead analyst).
- Automated integrity checks and metabolite name harmonization.
- Format of outputs is identical for each cohort analyses.
- Naming of output files is consistent for each cohort analyses.

https://bit.ly/3UYaF0Y

Load Data → Define Models → Run Models → Meta-Analyze

COMETS Analytics

1. Load Data – readCOMETSinput()

Data files from two cohorts cohort 1 and cohort 2:

```
# Retrieve the full paths of each input data
dir   <- system.file("extdata", package="RcometsAnalytics", mustWork=TRUE)
file1 <- file.path(dir, "cohort_1.xlsx")
file2 <- file.path(dir, "cohort_2.xlsx")

# Read in and process the input data
data1 <- RcometsAnalytics::readCOMETSinput(file1)
```

https://bit.ly/3UYaF0Y

```
data2 <- RcometsAnalytics::readCOMETSinput(file2)
```

1. Load Data – readCOMETSinput()

```
data2 <- RcometsAnalytics::readCOMETSinput(file2)
```

```
## VarMap sheet is read in.
## Metabolites sheet is read in.
## SubjectMetabolites sheet is read in.
## SubjectData sheet is read in.
## Models sheet is read in.
## Model_Types sheet is read in.
## There are 16 categorical variables.
## Running Integrity Check...
```

```
## Joining with `by = join_by(hmdb_id)`
```

```
## Begin testing models in Models sheet...
## Filtering subjects according to the rule(s) age< 70. 475 of 568 are retained.
```

```
## Warning in runModel.addRemVars(rem.obj, vars[oneVal], vars.type, "too few
## unique non-missing values", : The variable(s) female, fasted have been removed
## from adjvars because of: too few unique non-missing values
```

```
## Warning in runModel.addRemVars(rem.obj, tmp[rem], varSet, "correlated with
## another predictor", : The variable(s) multivitamin.2 have been removed from
## adjvars because of: correlated with another predictor
```

```
## Finished testing models in Models sheet.
```

https://bit.ly/3UYaF0Y

2. Get Model Data – getModelData()

Linear Regression model to run on both cohorts:

Metabolites ~ age + BMI

```
modeldata1 <- RcometsAnalytics::getModelData(data1, modelspec="Interactive",
                            adjvars="bmi_grp", outcomes=NULL, exposures="age")
modeldata2 <- RcometsAnalytics::getModelData(data2, modelspec="Interactive",
                            adjvars="bmi_grp", outcomes=NULL, exposures="age")
```

https://bit.ly/3UYaF0Y

3. Run models – runModel()

```
# Define names of files using convention described above
outfile1  <- paste0(getwd(), "/COMETStest/", "AgeAdjustedForBMI__cohort_1__", Sys.Date(), ".xlsx")
outfile2  <- paste0(getwd(), "/COMETStest/", "AgeAdjustedForBMI__cohort_2__", Sys.Date(), ".xlsx")

# Run the individual models with appropriate naming of cohorts:
results1  <- RcometsAnalytics::runModel(modeldata1, data1, "cohort_1",
             op=list(model="lm"))
results2  <- RcometsAnalytics::runModel(modeldata2, data2, "cohort_2",
             op=list(model="lm"))

# Write files locally using filenames defined above.
output1       <- RcometsAnalytics::OutputListToExcel(outfile1, results1)
```

```
## Output saved to file: /Users/mathee/Downloads/COMETStest/AgeAdjustedForBMI__cohort_1__2024-05-14.xls
x
```

```
output2       <- RcometsAnalytics::OutputListToExcel(outfile2, results2)
```

```
## Output saved to file: /Users/mathee/Downloads/COMETStest/AgeAdjustedForBMI__cohort_2__2024-05-14.xls
x
```

3. Run models – runModel() results

```
> names(results1)
[1] "ModelSummary"    "Effects"         "Errors_Warnings" "Table1"          "Info"
> head(results1$Effects,3)
  run           outcomespec exposurespec term     estimate   std.error  statistic     pvalue pvalue.adj estimate.lower estimate.upper
1   1              tyrosine          age  age -0.001444668 0.001996069 -0.7237564 0.46961158 0.88949714   -0.005356892    0.002467556
2   2 umbelliferone_sulfate          age  age  0.043084661 0.019531422  2.2059152 0.02792183 0.09307278    0.004803778    0.081365545
3   3        undecanedioate          age  age  0.002949424 0.011667204  0.2527961 0.80054742 0.88949714   -0.019917876    0.025816723
        metabolite_name
1              tyrosine
2 umbelliferone sulfate
3        undecanedioate
> head(results1$Table1)
  variable   in.model        type category   n n.unique min quartile1 median     mean quartile3 max n.missing
1      age   exposure  continuous          432       20  55        59     62 63.09259        67  74         0
2  bmi_grp adjustment categorical        1 141       NA  NA        NA     NA       NA        NA  NA        NA
3                                        2 164       NA  NA        NA     NA       NA        NA  NA        NA
4                                        3 122       NA  NA        NA     NA       NA        NA  NA        NA
5                                        4   5       NA  NA        NA     NA       NA        NA  NA        NA
```

4. Meta-analyze – runMeta()

Two output files running the same model on two different cohorts

outfile1 <- AgeAdjustedForBMI__cohort_1__2024-05-13.xlsx
outfile2 <- AgeAdjustedForBMI__cohort_2__2024-05-13.xlsx

```
mymeta <- RcometsAnalytics::runMeta(c(outfile1, outfile2))

# There are 3 sections for the results files:
names(mymeta)
```

```
## [1] "Metaresults"     "Errors_Warnings" "Info"
```

- **Metaresults**: harmonized metabolite name, exposure, number of cohorts used, estimates and p-values, etc.
- **Errors_Warnings**: warnings and errors associated with meta-analysis
- **Info**: date, R version, package version, model name, cohort names, input files, etc.

4. Meta-analyze – runMeta() results

```r
# View the first 3 rows of the results
head(mymeta$Metaresults, 3)
```

```
##                          outcome_uid exposure_uid term n.cohort n.sub
## 1                       CHEM100006282          age  age        2  1000
## 2 HMDB00158#HMDB00158_B1#HMDB00158_B2          age  age        2  1000
## 3                           HMDB00289          age  age        2  1000
##   fixed.pvalue fixed.estimate fixed.std.error random.pvalue random.estimate
## 1  0.008639827    0.033004821     0.012568542   0.008639827    0.0330048214
## 2  0.454284995    0.001013585     0.001354534   0.693471527    0.0008976198
## 3  0.020157598    0.003683725     0.001585488   0.085949308    0.0038438202
##   random.std.error het.pvalue cohort_1 cohort_2 cohort_1.estimate
## 1      0.012568542 0.50016623        1        1       0.043084661
## 2      0.002277365 0.09360158        1        1      -0.001444668
## 3      0.002238471 0.16130767        1        1       0.006254902
##   cohort_1.std.error cohort_1.run  cohort_1.outcomespec cohort_1.exposurespec
## 1        0.019531422            2 umbelliferone_sulfate                   age
## 2        0.001996069            1              tyrosine                   age
## 3        0.002425571            5                 urate                   age
##   cohort_2.estimate cohort_2.std.error cohort_2.run  cohort_2.outcomespec
## 1       0.025880721        0.016419959            2 umbelliferone_sulfate
## 2       0.003111862        0.001844140            1              tyrosine
## 3       0.001765603        0.002095011            5                 urate
##   cohort_2.exposurespec Duplicate
## 1                   age         0
## 2                   age         0
## 3                   age         0
```

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4. Meta-analyze – runMeta() results

```r
# View the first 3 rows of the results
head(mymeta$Metaresults, 3)
```

```
##                          outcome_uid exposure_uid term n.cohort n.sub
## 1                       CHEM100006282          age  age        2  1000
## 2 HMDB00158#HMDB00158_B1#HMDB00158_B2          age  age        2  1000
## 3                            HMDB00289          age  age        2  1000
##   fixed.pvalue fixed.estimate fixed.std.error random.pvalue random.estimate
## 1  0.008639827    0.033004821     0.012568542   0.008639827    0.0330048214
## 2  0.454284995    0.001013585     0.001354534   0.693471527    0.0008976198
## 3  0.020157598    0.003683725     0.001585488   0.085949308    0.0038438202
##   random.std.error het.pvalue cohort_1 cohort_2 cohort_1.estimate
## 1      0.012568542 0.50016623        1        1       0.043084661
## 2      0.002277365 0.09360158        1        1      -0.001444668
## 3      0.002238471 0.16130767        1        1       0.006254902
##   cohort_1.std.error cohort_1.run  cohort_1.outcomespec cohort_1.exposurespec
## 1        0.019531422            2 umbelliferone_sulfate                   age
## 2        0.001996069            1              tyrosine                   age
## 3        0.002425571            5                 urate                   age
##   cohort_2.estimate cohort_2.std.error cohort_2.run  cohort_2.outcomespec
## 1       0.025880721        0.016419959            2 umbelliferone_sulfate
## 2       0.003111862        0.001844140            1              tyrosine
## 3       0.001765603        0.002095011            5                 urate
##   cohort_2.exposurespec Duplicate
## 1                   age         0
## 2                   age         0
## 3                   age         0
```

Harmonized Metabolite Name

Estimates for Fixed-effects and random effects (DerSimonian-Laird Method)

Cochran's Q-test for heterogeneity

Whether or not duplicate metabolites were identified – currently they would be removed

5. Meta-analyze in “super-batch” mode– runAllMeta()

Given a set of input folders with multiple cohort results, the function will identify same models run on multiple cohorts and will perform meta-analyses.

3 arguments can be input:

1. **filesFolders**: is a character vector of file names and/or folder names.
2. **out.dir**: output folder to write the results to
3. **opfile**: optional file for meta-analysis options.

Here, we simply input the working folder where we saved our previous runModel() results:

```
data.dir <- paste0(getwd(),"/COMETStest")
allmeta_results <- RcometsAnalytics::runAllMeta(data.dir, data.dir)
```

5. Meta-analyze in “super-batch” mode– runAllMeta()

”Super-batch” result

```
## Begin meta-analysis for model 'AgeAdjustedForBMI'
## $Metaresults
##                          outcome_uid exposure_uid term n.cohort n.sub
## 1                       CHEM100006282          age  age        2  1000
## 2  HMDB00158#HMDB00158_B1#HMDB00158_B2          age  age        2  1000
## 3                            HMDB00289          age  age        2  1000
## 4                            HMDB00294          age  age        2  1000
## 5               HMDB00296#HMDB00296_B1          age  age        2  1000
## 6                            HMDB00300          age  age        2  1000
## 7                            HMDB00883          age  age        2  1000
## 8                            HMDB00888          age  age        2  1000
## 9                            HMDB00946          age  age        2  1000
## 10                           HMDB13128          age  age        2  1000
##    fixed.pvalue fixed.estimate fixed.std.error random.pvalue random.estimate
## 1  8.639827e-03    0.033004821     0.012568542  8.639827e-03    0.0330048214
## 2  4.542850e-01    0.001013585     0.001354534  6.934715e-01    0.0008976198
## 3  2.015760e-02    0.003683725     0.001585488  8.594931e-02    0.0038438202
## 4  5.783638e-06    0.009860259     0.002174666  5.783638e-06    0.0098602589
## 5  1.356218e-01   -0.002387080     0.001599603  4.725153e-01   -0.0
## 6  1.333316e-01   -0.013402300     0.008928363  1.333316e-01   -0.0
## 7  6.769593e-02   -0.001908100     0.001044375  6.769593e-02   -0.0
## 8  8.605485e-01    0.001407602     0.008012491  8.605485e-01    0.0
## 9  7.203521e-01   -0.008061107     0.022517794  7.203521e-01   -0.0080611068
## 10 2.559844e-02    0.017988411     0.008058421  3.229090e-01    0.0169516261
##    random.std.error het.pvalue cohort_1 cohort_2 cohort_1.estimate
## 1       0.012568542 0.50016623        1        1       0.043084661
## 2       0.002277365 0.09360158        1        1      -0.001444668
## 3       0.002238471 0.16130767        1        1       0.006254902
## 4       0.002174666 0.91232983        1        1       0.010147079
## 5       0.002775199 0.08938936        1        1       0.000998012
## 6       0.008928363 0.54986083        1        1      -0.007318610
## 7       0.001044375 0.36349346        1        1      -0.002880340
## 8       0.008012491 0.85573824        1        1       0.002949424
## 9       0.022517794 0.92255525        1        1      -0.010624082
```

…

runMeta() result

```
# View the first 3 rows of the results
head(mymeta$Metaresults, 3)
```

```
##                          outcome_uid exposure_uid term n.cohort n.sub
## 1                       CHEM100006282          age  age        2  1000
## 2 HMDB00158#HMDB00158_B1#HMDB00158_B2          age  age        2  1000
## 3                            HMDB00289          age  age        2  1000
##   fixed.pvalue fixed.estimate fixed.std.error random.pvalue random.estimate
## 1  0.008639827    0.033004821     0.012568542   0.008639827    0.0330048214
## 2  0.454284995    0.001013585     0.001354534   0.693471527    0.0008976198
## 3  0.020157598    0.003683725     0.001585488   0.085949308    0.0038438202
##   random.std.error het.pvalue cohort_1 cohort_2 cohort_1.estimate
## 1      0.012568542 0.50016623        1        1       0.043084661
## 2      0.002277365 0.09360158        1        1      -0.001444668
## 3      0.002238471 0.16130767        1        1       0.006254902
```

Results are identical!

```
   t_1.std.error cohort_1.run  cohort_1.outcomespec cohort_1.exposurespec
## 1        0.019531422            2 umbelliferone_sulfate                  age
## 2        0.001996069            1              tyrosine                  age
## 3        0.002425571            5                 urate                  age
##   cohort_2.estimate cohort_2.std.error cohort_2.run  cohort_2.outcomespec
## 1       0.025880721        0.016419959            2 umbelliferone_sulfate
## 2       0.003111862        0.001844140            1              tyrosine
## 3       0.001765603        0.002095011            5                 urate
##   cohort_2.exposurespec Duplicate
## 1                   age         0
## 2                   age         0
## 3                   age         0
```

COMETS Analytics

COMETS Analytics Enhances Rigor and Reproducibility, thereby accelerating discoveries

Standardized Input (with integrity checks and harmonization)

Standardized Output (with harmonized metabolites)

Miminizes burden on data analysts, avoids data transfer, enables streamlined process:

Load Data → Define Models → Run Models → Meta-Analyze

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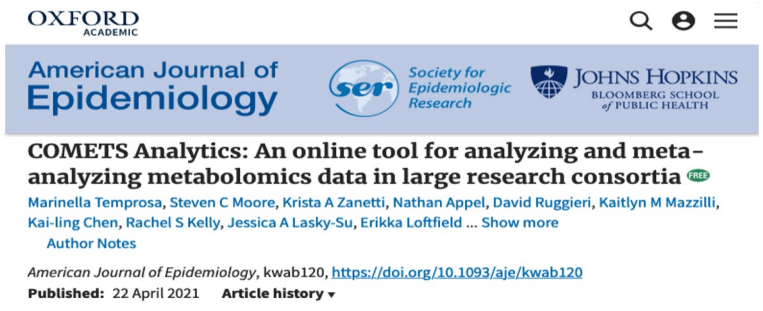
Users and other COMETS Interest Groups!
Thank you for your feedback!

Contact us at comets.analytics@gmail.com

COMETS Analytics

Next …Discussion /Wrap-Up



COMETS Analytics Public and Secure Web Interface

https://bit.ly/3wtd0aQ

Example Code and Documents

https://bit.ly/NIH_NCATS_Met_Epi_workshop

COMETS Analytics R Package and Documentation

https://bit.ly/4biAXAz